Volume V · Chapter 25 — Learning for Control (Imitation/RL, sim-to-real) · Theory

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-V-Chapter-25-context.md` for the AI interaction log.

Prerequisites: Ch 23, Ch 24.

Learning outcomes — the student can: explain imitation learning & RL for control; train a policy in simulation; reason about sim-to-real transfer and its risks; apply safety constraints/shields.

Topics (theory) — 10 h:

#	Topic	h
25.1	Control paradigms: classical vs. learned	1
25.2	Imitation learning (behavioral cloning)	2
25.3	Reinforcement learning basics for control	3
25.4	Sim-to-real transfer: domain randomization, gaps	2
25.5	Safety constraints & constrained policies	2

Datasets/tools: RL libraries (Stable-Baselines3 / Isaac Gym); the simulator. **Assessment:** trained policy (**60%**); sim-to-real analysis (**20%**); quiz (**20%**). **Key**

decisions: IL vs. RL; reward design; safety shielding; when sim is insufficient. **References:** plan §7; §13 → *Robotics & embodied AI*. **Hours:** Theory **10** + Lab **14** = **24**.